Machine Learning

Credit Card Fraud Detection (ML_CreditCard.ipynb)

1) Overview

This project builds a machine learning model to detect **fraudulent credit card transactions**. Fraud detection is a **highly imbalanced binary classification** task, where fraudulent transactions are very rare compared to legitimate ones. Because of this imbalance, evaluation should focus on metrics like **precision, recall, and F1-score** (especially for the fraud class), not just accuracy.

Target column: `Class`

- `0` → Legitimate transaction
- `1` → Fraudulent transaction

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2) Dataset Summary

The notebook uses a credit card transactions dataset with anonymized features.

- **Dataset shape used in notebook: (27,819, 31)**
- **Columns:** `Time`, `Amount`, `V1` … `V28`, `Class`
- **Missing labels:** rows with NaN in `Class` were checked/dropped → **0 rows dropped** (as per notebook output)

Class distribution (Imbalance)

- **Fraud (Class = 1): 93**
- **Legitimate (Class = 0): 27,725**
- **Fraud fraction (outlierFraction): 0.00335** (~0.335%)

This confirms extreme imbalance: the model can achieve high accuracy by predicting “legitimate” most of the time, so accuracy alone is not sufficient.

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3) Exploratory Data Analysis (EDA)

3.1 Amount vs Class

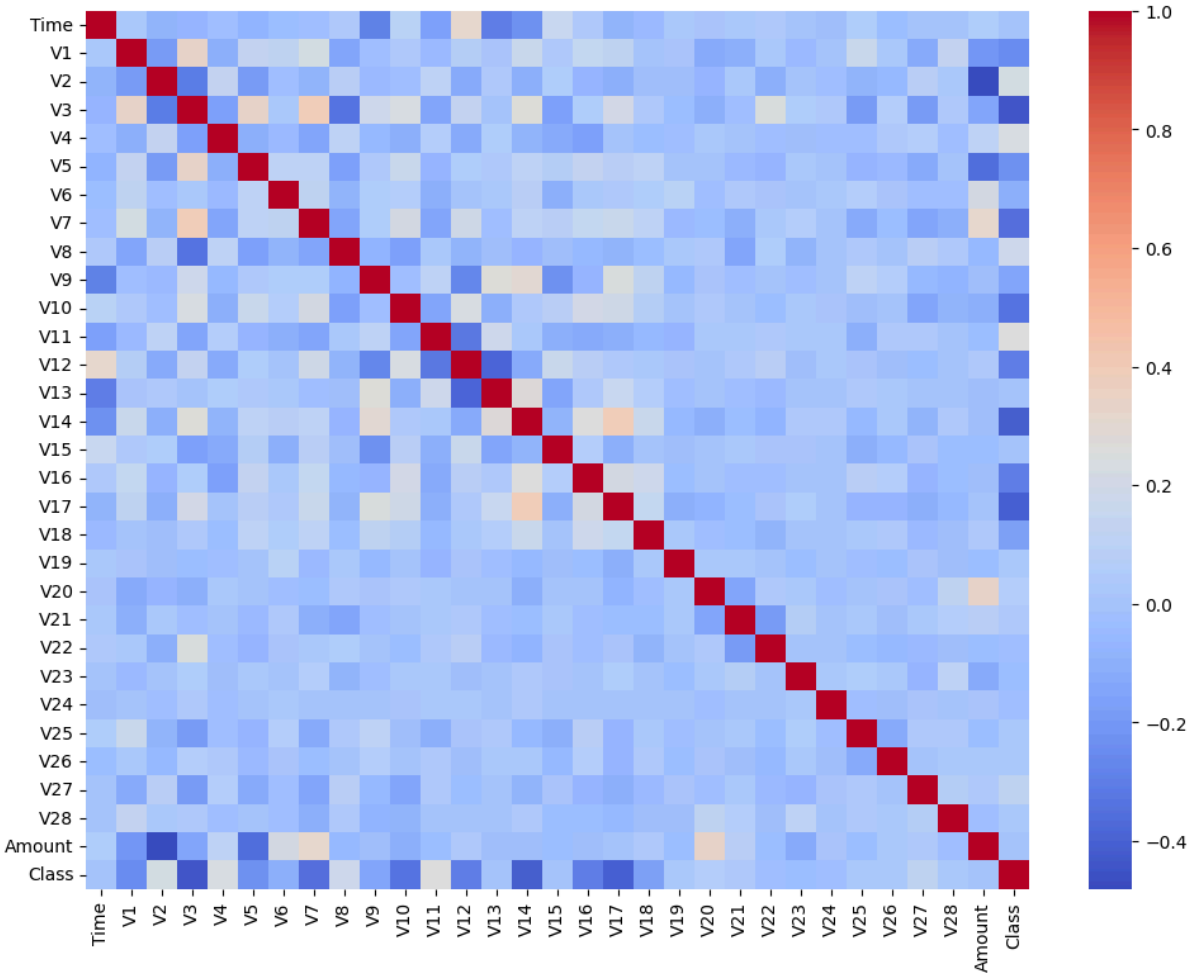
The notebook visualizes transaction amounts for fraud vs legitimate transactions (e.g., histogram/plot).

Insights (based on the plot + describe stats):

- Fraud transactions are rare across all amount ranges.
- The `Amount` distribution differs between fraud and legitimate transactions, which can provide signal for classification.
- Summary statistics (`describe()`) were computed separately for fraud and legitimate transactions to compare typical amounts and spread.

3.2 Correlation Heatmap

A correlation heatmap is plotted to understand relationships between variables.



Insights:

- Many anonymized `V` features are weakly correlated with each other and with `Class`, which is expected for transformed/anonymized features.
- Potentially informative features can still be captured by non-linear models like Random Forest.

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4) Data Preparation

4.1 Feature/Label Split

- **X (features):** all columns except `Class`
- **y (label):** `Class`

4.2 Train/Test Split

The dataset was split into training and testing sets using scikit-learn:

```
x_train, x_test, y_train, y_test = train_test_split(xd, yd,
test_size=0.2, random_state=42)
```

- **Training:** 80%
- **Testing:** 20%
- **random_state=42** ensures reproducibility.

> Note: Because this is imbalanced data, using `stratify=y` is often recommended to preserve class distribution across splits. (Confirm whether you added stratification; if not, it can be a future improvement.)

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5) Model Training

A **Random Forest Classifier** is trained as the baseline model.

Why Random Forest?

- Works well on tabular datasets
- Captures non-linear patterns
- Provides a strong baseline with minimal preprocessing

Model details (fill if you set hyperparameters explicitly):

- `n_estimators = _____`
- `max_depth = _____`
- `class_weight = _____` (if used)

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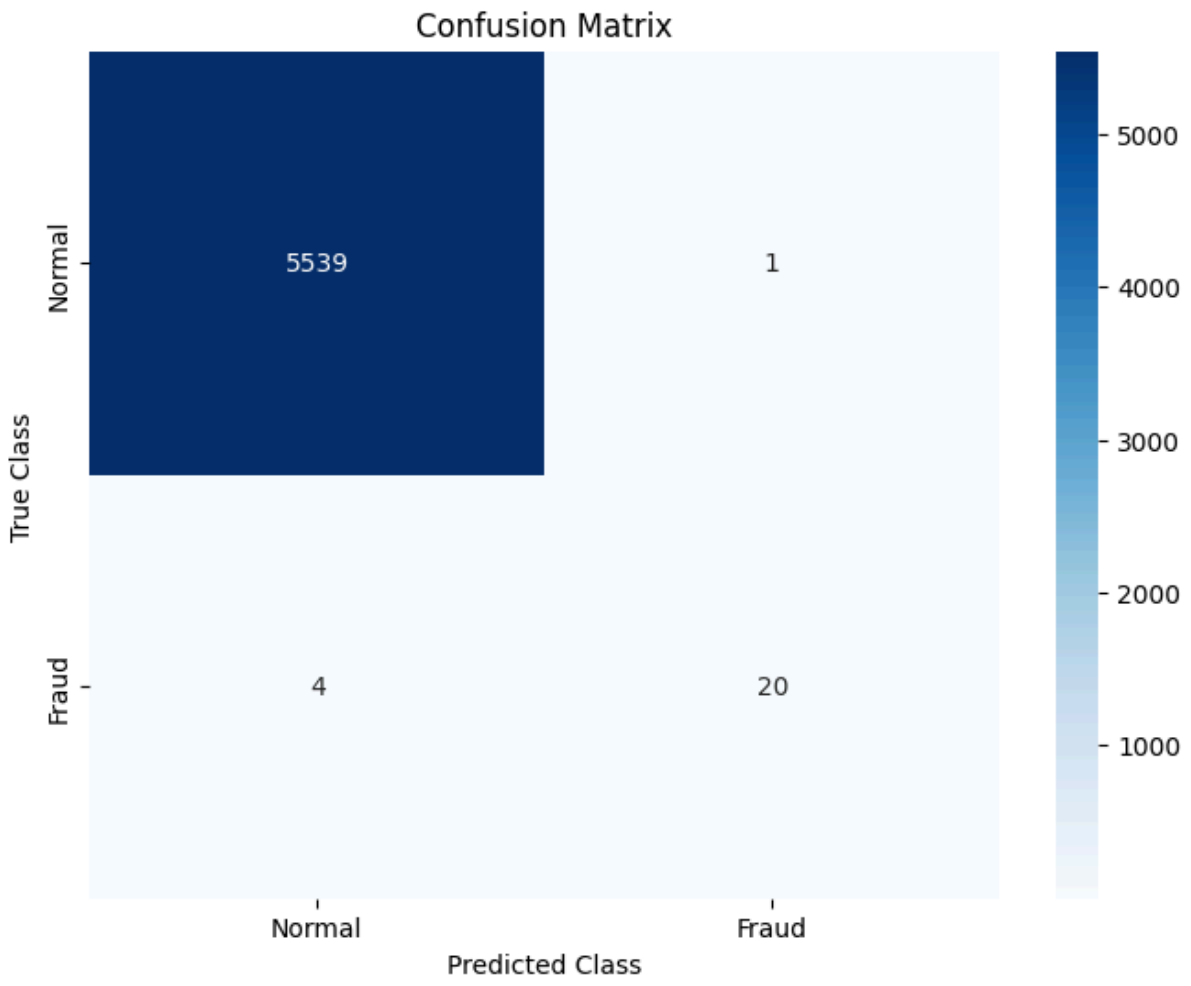
6) Model Evaluation (Most Important Section)

Since fraud detection is imbalanced, evaluation focuses on fraud-class performance.

6.1 Confusion Matrix

Paste your confusion matrix output here:



- **TP:** fraud correctly detected
- **FP:** legitimate incorrectly flagged as fraud

- **FN:** fraud missed (most costly)
- **TN:** legitimate correctly identified

6.2 Classification Report

Paste your `classification_report(y_test, y_pred)` here:

```
              precision    recall  f1-score   support

         0.0       1.00      1.00      1.00      3566
         1.0       0.94      0.83      0.88        18

    accuracy                           1.00      3584
   macro avg       0.97      0.92      0.94      3584
weighted avg       1.00      1.00      1.00      3584
```

- Fraud Precision (Class=1): _____
- Fraud Recall (Class=1): _____
- Fraud F1-score (Class=1): _____
- Accuracy: _____ *(include but don't treat as the main metric)*

6.3 Interpretation

- **High recall** means the model catches more fraud (fewer frauds missed).
- **High precision** means fewer legitimate transactions are incorrectly blocked.
- The best balance depends on real-world cost tradeoffs (fraud loss vs customer friction).

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7) Results Summary Table

Fill in using your metrics:

Model	Fraud Precision	Fraud Recall	Fraud F1	Notes
Random Forest	____	____	____	Baseline